

WASP-135b

R-0196

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-135_250816 · created 2026-07-16T21:36:50+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2460903.7461 +/- 0.0039 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1754 +/- 0.0055
Transit depth (Rp/Rs)^2	3.08 +/- 0.19 [%]
Orbital Inclination (inc)	81.86 +/- 1.68
Airmass coefficient 1 (a1)	0.653 +/- 0.048
Airmass coefficient 2 (a2)	0.384 +/- 0.08
Scatter in the residuals of the lightcurve fit is	7.38 %
Best Comparison Star	#1 - [381, 398]
Optimal Aperture	2.46
Optimal Annulus	13.29
Transit Duration (day)	0.06 +/- 0.01

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.1754 ± 0.0055 vs 0.139 (deviation 2.05σ)
Duration fit vs published	0.06 d vs 0.0687 d (deviation 0.27σ)
Mid-time O–C	-36.9 min
Depth SNR	9.9
Residual scatter	7.38 %

Light curve

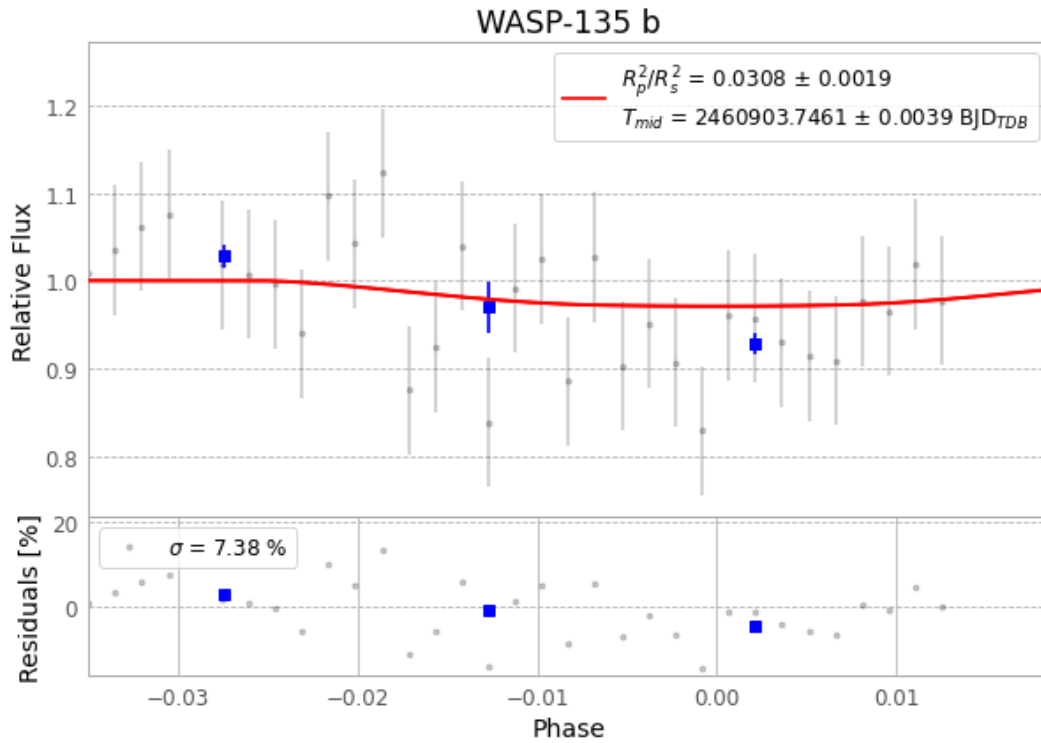


Figure 1 — FinalLightCurve_WASP-135 b_2025-08-15.png (SHA-256 b8abbe3757ae49e2...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [283, 269], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 b6add2e09a846d5f...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1965f5e

Data provenance

38 FITS frames (25 MB), WASP-135250816043915.FITS → WASP-135250816063015.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-135-250816.log (a3670c0c7785...), FinalLightCurve_WASP-135 b_2025-08-15.png (b8abbe3757ae...), FinalLightCurve_WASP-135 b_2025-08-15.csv (d6c9c1e012d4...)

Compute resources

Wall time 145.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-16 21:36Z.